

MATH 213 – Homework 4

Reading. Before beginning the written questions,

- Read the rest of section 2.2, and section 2.4, the chapter summary.

Written homework

As always, please type your answers directly into this file, SAVE the file as a PDF and then upload it to Blackboard for grading.

1. Questions on random sampling topics that we did not discuss in class, described in section 2.1. Turn in your responses to these questions:

- a. “A man on the news got mercury poisoning from eating swordfish, so the average mercury concentration in swordfish must be dangerously high.” What is this type of evidence called? According to the textbook, what is the problem with drawing the conclusion that the average mercury content must be high from this type of evidence?

This type of evidence is called anecdotal evidence, the issue with drawing a conclusion that the average mercury content must be high from this type of evidence is that anecdotal evidence is based on a single or small number of observations which may not be representative of the entire population. To be more specific to this problem it does not provide a reliable or generalizable measure of the average mercury concentration in swordfish.

- b. In class, we talked about using a computer (we would use R) to generate a simple random sample. In this kind of sampling, every member of the population has an equal likelihood of being selected to be in the sample. This is the main kind of sampling method we will focus on in this course, but our book does mention other sampling methods. List them here with a brief (1-sentence) description.

- 1) **Cluster sampling:** A sampling technique used in statistics where the entire population is divided into separate groups, or clusters, that are each a mini representation of the entire population.
- 2) **Convenience sample:** sampling method where the sample is taken from a group of people easy to contact or to reach.
- 3) **Multistage sample:** Like cluster sampling. However, instead of sampling every element within a selected cluster, a further stage of sampling is carried out within each of these clusters. This could involve selecting sub-clusters or even individual elements within each cluster through additional sampling methods.
- 4) **Stratified sampling:** A sampling technique where the researcher divides the entire population into different subgroups or strata, based on shared characteristics or attributes. Each group/stratum is then sampled proportionally.

- c. Describe the difference between simple random sampling and multistage sampling, according to our textbook

Simple random sampling: Where every member of the population has a fair and equal chance of being selected, and the sample is drawn directly from the entire population.

Multistage sampling: The sampling process occurs in multiple stages, normally combining different sampling methods and selecting individuals within those stages.

2. Another question on observational study vocabulary we did not discuss in class: Write at least a sentence describing the distinction between the two types of observational studies mentioned in the textbook.

Prospective study: Researchers identify individuals and collect data as events unfold over time.

Retrospective study: Researchers collect data after events have occurred, usually by reviewing past records.

3. Write at least one sentence describing why observational studies can (usually) only show associations, not causal associations.

Observational studies can only show associations because researchers do not control the variables or assign treatments. There may be confounding variables that influence the observed relationship making it difficult to establish causation.

4. Write at least one sentence describing why experiments *do not* have that limitation, and *can* show causal associations.

Experiments can show causal associations because researchers actively control and manipulate variables, randomly assign treatments and account for confounding factors. This allows for the establishment of cause and effect relationships.

5. The chapter review contains a helpful diagram that tries to summarize the differences between the two separate ways we use randomization in study design: random sampling and random allocation. Write a few sentences describing what random sampling is and why it is useful, and what random allocation is and why *it is* useful.

Random sampling is the process of selecting a sample from a population where every individual has a fair and equal chance of being selected for a group. It is useful because it ensures that the sample is representative of the population, reducing bias, and allows researchers to generalize results after an experiment has concluded. Similarly **random allocation** is the process of assigning individuals in a study to different treatment groups randomly. It is useful because it helps eliminate bias, balances confounding variables across groups and allows for causal inferences to be made.

Lastly, do Question 30 CH2

30. **Screens, teens, and psychological well-being.** In a study of three nationally representative large-scale datasets from Ireland, the United States, and the United Kingdom ($n = 17,247$), teenagers between the ages of 12 to 15 were asked to keep a diary of their screen time and answer questions about how they felt or acted. The answers to these questions were then used to compute a psychological well-being score. Additional data were collected and included in the analysis, such as each child's sex and age, and on the mother's education, ethnicity, psychological distress, and employment. The study concluded that there is little clear-cut evidence that screen time decreases adolescent well-being. ([Orben and Baukney-Przybylski 2018](#))

- a. What type of study is this?
observational
- b. Identify the explanatory variables.

Screen time: the diary entries where the teens recorded their screen time

- c. Identify the response variable.

Psychological wellbeing score computed from the teen's answers

- d. Comment on whether the results of the study can be generalized to the population, and why.

The study can be generalized because they used the nationally representative large-scale datasets from Ireland, the United States and United Kingdom the results are generalizable to the population of adolescents (12-15) within those countries.

- e. Comment on whether the results of the study can be used to establish causal relationships.

No, the results of this study cannot establish a causal relationship because the study itself is observational and correlational in nature. It only defines associations between screen time and psychological wellbeing. There is no experimental manipulation or random assignment, and potential confounding variables may influence the observed relationships.